Module 7. Neural Networks

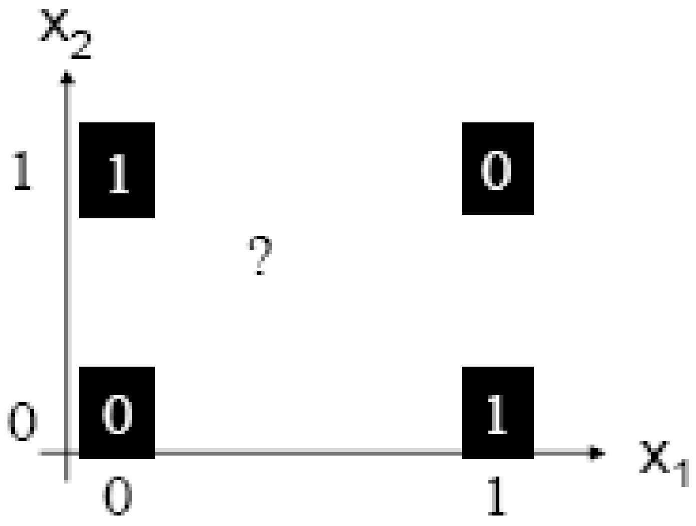
- Basic Concepts
- Perceptron Learning
- Artificial Neural Networks
- Backpropagation
- Summary

The Exclusive OR problem

A Perceptron cannot represent Exclusive OR since it is not linearly separable.

XOR function

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0



Solution to Non-Linearly Separatable Problem?

- So far, we have used $f(\mathbf{x},\mathbf{w}) = sign(\sum_{1=1}^{n} w_i x_i)$
- We could preprocess the inputs in a non-linear way $f(\mathbf{x},\mathbf{w}) = sign(\sum_{1=1}^{m} w_i \phi_i(\mathbf{x}))$
- To the perceptron algorithm it looks just the same and can use the same learning algorithm, it just has different inputs - SVM
- For example, for a problem with two inputs *x* and *y* (plus the bias), we could also add the inputs x^2, y^2, and $x \cdot y$
- The perceptron would just think it is a 5 dimensional task, and it is linear in those 5 dimensions
    - But what kind of decision surfaces would it allow for the original 2-*d* input space?

Quadric Machine

- All quadric surfaces (2nd order)
  - ellipsoid
  - parabola
  - etc.
- That significantly increases the number of problems that can be solved
- But still many problem which are not quadrically separable
- Could go to 3rd and higher order features, but number of possible features grows exponentially